

ROADRESQ

Software Requirements Specification

Version 1.0 • Production-Oriented Requirements Baseline

Product: Digital Roadside Assistance & Automotive Service Platform

Target stack: Next.js / React • FastAPI • PostgreSQL/PostGIS • Redis • WebSockets

AUTH → REQUEST → MATCH → TRACK → JOB → BILL → PAY → REVIEW

Document Control

Field	Value
Document	Software Requirements Specification (SRS)
Product	RoadResQ
Version	1.0
Status	Engineering baseline
Audience	Product, frontend, backend, QA, DevOps, security and operations

Revision History

Version	Date	Description
1.0	04 Sep 2026	Initial production-oriented SRS baseline

1. Introduction

1.1 Purpose

This document defines the functional, non-functional, integration, data, security, operational and quality requirements for RoadResQ. It is the engineering baseline for implementing and validating the platform.

1.2 Product Scope

RoadResQ connects vehicle owners with verified roadside assistance providers, mechanics, garages, towing, battery, tyre, fuel, mechanical and scheduled-service providers. The platform supports the journey from request through matching, live progress, inspection, estimate approval, repair completion, billing, payment and review.

1.3 Goals

- Reduce the time required to find appropriate vehicle assistance.
- Provide transparent provider identity, service capability, availability and ETA.
- Digitize the service lifecycle instead of relying on calls, WhatsApp or paper records.
- Give providers operational tools for requests, jobs, estimates, completion and earnings.
- Give administrators control over verification, support, disputes, payments and operations.
- Create an extensible foundation for garages, fleets, parts, insurance, EV services and AI.

1.4 MVP Out of Scope

- Replacement of police, ambulance or fire emergency services.
- Full insurance claim processing.
- Large-scale parts marketplace.
- Advanced AI diagnosis and predictive maintenance.

2. Product Overview

2.1 User Classes

Role	Description	Core capabilities
Customer	Vehicle owner	Vehicles, requests, tracking, approvals, payments, history, review

Role	Description	Core capabilities
Service Provider	Mechanic, garage, towing or roadside assistance	Verification, availability, requests, jobs, estimates, completion, etc.
Admin / Operations	Platform operator	Verification, users, bookings, payments, disputes, support, analytics
Support Agent	Support role	Tickets, communication and controlled customer/job access

2.2 Service Categories

- Mechanical assistance
- Towing
- Battery assistance/replacement
- Tyre puncture/change
- Fuel delivery
- Accident/vehicle recovery
- General roadside assistance
- Scheduled garage service

2.3 Core Journey

Request → Match → Accept → Track → Arrive → Inspect → Approve → Repair → Pay → Review

3. Functional Requirements

3.1 Authentication & Authorization

ID	Requirement
FR-AUTH-001	Users shall be able to register using supported identity verification.
FR-AUTH-002	System shall authenticate users and issue secure sessions/access credentials.
FR-AUTH-003	System shall support OTP verification where configured.
FR-AUTH-004	System shall support expiry, refresh and logout of sessions/tokens.
FR-AUTH-005	RBAC shall enforce customer, provider, support and admin permissions.
FR-AUTH-006	Accounts shall support active, suspended, blocked and pending states.
FR-AUTH-007	Protected resources shall enforce authentication plus object-level authorization.

3.2 Customer & Vehicle Management

ID	Requirement
FR-CUST-001	Customer shall view and update permitted profile fields.
FR-CUST-002	Customer shall add, edit, deactivate and select vehicles.
FR-CUST-003	Vehicle records shall support registration/identifier, make, model, variant and fuel/EV type.
FR-CUST-004	Customer shall view service history, estimates, invoices and completed jobs.
FR-CUST-005	Customer shall search/filter service providers by category, location, availability and quality signals.

3.3 Assistance Request

ID	Requirement
FR-REQ-001	Customer shall create an assistance request for a selected vehicle.
FR-REQ-002	Customer shall select a problem/service category.
FR-REQ-003	System shall capture GPS location when permitted and allow manual location.
FR-REQ-004	Customer may attach description, photos, video or voice evidence where supported.
FR-REQ-005	Customer shall see request state and important timestamps.
FR-REQ-006	Customer shall cancel when permitted by booking state/policy.
FR-REQ-007	System shall prevent accidental duplicate active requests.

3.4 Provider Matching & Dispatch

ID	Requirement
FR-MATCH-001	Matching shall consider provider status, availability, service capability and vehicle/service compatibility.
FR-MATCH-002	Distance/ETA shall be used as a matching signal.
FR-MATCH-003	Rating, acceptance/completion performance and configurable business rules may affect ranking.
FR-MATCH-004	Eligible providers shall receive dispatch requests.
FR-MATCH-005	Unanswered dispatches shall timeout/retry according to configurable policy.
FR-MATCH-006	Assignment shall be transactionally protected so one request cannot be claimed by multiple providers.
FR-MATCH-007	System shall retry or present a no-provider state when matching fails.

3.5 Provider Registration & Verification

ID	Requirement
FR-PROV-001	Provider shall register business/profile information.
FR-PROV-002	Provider shall securely upload required verification documents.
FR-PROV-003	Admin shall approve, reject or request changes.
FR-PROV-004	Provider shall configure offered services and supported vehicle categories.
FR-PROV-005	Provider shall control online/offline availability and hours.
FR-PROV-006	Provider may configure service radius.
FR-PROV-007	Admin shall suspend/reactivate providers subject to policy.

3.6 Booking & Job Lifecycle

The server shall enforce the state machine: **REQUESTED** → **SEARCHING** → **PROVIDER_ASSIGNED** → **ACCEPTED** → **ON_THE_WAY** → **ARRIVED** → **INSPECTION** → **ESTIMATE_PENDING** → **APPROVED** → **IN_PROGRESS** → **COMPLETED** → **INVOICE_GENERATED** → **PAYMENT_COMPLETED** → **CLOSED**. Exception states include **CANCELLED**, **EXPIRED**, **DISPUTED**.

ID	Requirement
FR-JOB-001	Accepted bookings shall have one active provider assignment.
FR-JOB-002	System shall support arrival verification such as OTP.
FR-JOB-003	Provider shall record inspection findings and evidence.
FR-JOB-004	Provider shall create structured estimates with line items, prices and taxes/fees.

ID	Requirement
FR-JOB-005	Customer approval shall be required before estimate-controlled work.
FR-JOB-006	Additional work shall require updated approval where policy requires.
FR-JOB-007	Provider shall record completion details.
FR-JOB-008	Important state transitions shall be timestamped and attributable.

3.7 Live Tracking

ID	Requirement
FR-TRACK-001	System shall receive provider location updates during active tracking.
FR-TRACK-002	Customer shall see provider progress and ETA when mapping data permits.
FR-TRACK-003	Tracking shall start/end according to booking state and privacy policy.
FR-TRACK-004	Stale/unavailable location shall produce a graceful fallback.
FR-TRACK-005	Location shall be accessible only to authorized participants and operational systems.

3.8 Billing, Payments & Reviews

ID	Requirement
FR-BILL-001	System shall generate structured invoices from completed/approved charges.
FR-BILL-002	System shall record payment status, amount, method and gateway reference.
FR-BILL-003	MVP shall support UPI and cash recording, with architecture for future methods.
FR-BILL-004	Payment operations shall be idempotent.
FR-BILL-005	Payment/payout records shall support reconciliation.
FR-BILL-006	Eligible completed jobs shall support customer reviews.
FR-BILL-007	Provider earnings shall be represented separately from customer payment.

3.9 Notifications & Communication

- Notify customers for request creation, provider found, acceptance, arrival, estimate, completion, invoice, payment and review.
- Notify providers for new requests, cancellations, estimate approvals and relevant job events.
- Support configurable push/SMS/email adapters.
- Support chat/masked calling for eligible active bookings where implemented.
- Record notification delivery state and prevent avoidable duplicates.

3.10 Admin & Operations

ID	Requirement
FR-ADM-001	Admin shall view operational KPIs and active job states.
FR-ADM-002	Admin shall search and manage user accounts subject to permissions.
FR-ADM-003	Admin shall verify providers and manage provider status.
FR-ADM-004	Admin shall monitor active, completed, cancelled and disputed jobs.
FR-ADM-005	Support roles shall manage support tickets and notes.

ID	Requirement
FR-ADM-006	System shall support controlled dispute review/resolution.
FR-ADM-007	Sensitive administrative actions shall be auditable.

4. Non-Functional Requirements

ID	Requirement
NFR-PERF-001	Common APIs shall target low latency under normal production load; exact SLOs shall be finalized
NFR-PERF-002	Realtime status/tracking shall be resilient with graceful fallback.
NFR-SCALE-001	Stateless application components shall support horizontal scaling.
NFR-AVAIL-001	Production services shall use health checks and recovery mechanisms.
NFR-SEC-001	Production traffic shall use TLS; secrets shall not be hard-coded.
NFR-SEC-002	RBAC and object-level authorization shall be enforced server-side.
NFR-SEC-003	Inputs/uploads shall be validated and constrained.
NFR-DATA-001	Transactions, constraints and relationships shall preserve data integrity.
NFR-PRIV-001	Personal, location, payment and provider-document data shall be minimized and access-controlled
NFR-OBS-001	Critical failures shall be observable through structured logs, metrics and alerts.
NFR-REC-001	Backups and recovery shall be tested against defined RPO/RTO targets.
NFR-UX-001	Critical flows shall be responsive and usable on mobile.
NFR-ACC-001	UI shall target recognized accessibility best practices.
NFR-MAINT-001	Code shall use modular boundaries, documented contracts and automated tests.

5. External Interface Requirements

5.1 User Interfaces

- Customer: home, request, provider results, booking details, live tracking, estimate approval, invoice/payment, vehicles, history and profile.
- Provider: onboarding, verification, availability, request inbox, active job, inspection, estimate, completion and earnings.
- Admin: dashboard, users, providers, bookings, payments, support, disputes and audit views.
- All interfaces shall define loading, empty, error, offline, denied and success states.

5.2 Maps & Geolocation

A mapping provider shall support map display, geocoding, routing and ETA. Integration shall be isolated behind an internal adapter to allow provider replacement.

5.3 Payments

Payment integration shall support secure initiation, callback/webhook verification, idempotency, reconciliation and failure handling.

5.4 Notifications

Push, SMS and email shall be implemented as replaceable adapters rather than coupling notification vendors to booking logic.

5.5 File Storage

Photos and documents shall use object storage or equivalent; large binary payloads should not be stored directly in core relational tables.

6. Data Requirements

6.1 Core Entities

Entity	Purpose
users	Identity, account and role information
vehicles	Customer-owned vehicles
providers	Provider business/profile/status
provider_documents	Verification documents and review state
provider_services	Provider capabilities
provider_availability	Operating/online availability
bookings	Primary assistance/service transaction
booking_locations	Request and operational locations
booking_status_history	State transition history
provider_location_updates	Tracking data
job_cards	Operational work record
inspections / inspection_items	Inspection findings
estimates / estimate_items	Proposed charges and approvals
invoices / invoice_items	Final billing
payments / payouts	Money movement and settlement
reviews	Customer feedback
notifications	Notification history
messages	Booking communication
support_tickets	Support/dispute workflows
audit_logs	Security/admin audit trail

6.2 Data Integrity Rules

- Vehicle ownership must be checked before customer operations on a vehicle.
- Only eligible providers may receive/accept a request.
- Booking transitions must be validated against the state machine.
- Financial records should be immutable or versioned after finalization.
- Critical operations shall use transaction boundaries and unique/idempotency constraints.

7. Business Rules

- A provider cannot accept a request unless active, eligible and available.
- A booking can have only one active provider assignment.

- Customer approval is required for estimate-controlled work.
- Provider location is restricted to active operational contexts.
- Cancellation eligibility/fees depend on configurable policy and booking state.
- Reviews are allowed only for eligible completed jobs and should prevent duplicates.
- Admin actions affecting verification, disputes, payments or account status require permission and audit logging.
- RoadResQ assistance UX must not imply replacement of public emergency services.

8. Security & Privacy Requirements

- Use secure authentication, least privilege, RBAC and object-level authorization.
- Use modern password hashing if passwords are supported; protect OTPs and session credentials.
- Never log passwords, OTPs, access tokens, API secrets or payment secrets.
- Validate upload type, size and content; restrict storage access.
- Rate-limit authentication, OTP, request creation and abuse-sensitive endpoints.
- Use TLS, secure headers, appropriate CORS/CSRF protections and safe cookie/token practices.
- Encrypt sensitive data appropriately and manage keys/secrets securely.
- Define retention/deletion policies for location, contact, vehicle, payment and provider-document data.
- Audit administrative and security-sensitive actions.

9. Reliability, Failure Handling & Edge Cases

Scenario	Expected behavior
No provider	Show clear no-match state and allow retry/alternative.
Provider rejects	Return request to eligible dispatch pool.
Provider cancels	Record cancellation and attempt reassignment where policy permits.
GPS unavailable	Allow manual location and communicate ETA limitations.
Network loss	Persist server state, retry safely and prevent duplicates.
Payment timeout	Do not assume success; reconcile using gateway status/webhook.
Duplicate webhook	Process idempotently.
Estimate rejected	Keep controlled pending/requote/cancel state per policy.
Server/app crash	Recover authoritative state from persisted records.
Dispute	Route to authorized support/admin workflow and flag relevant records.

10. API & Integration Requirements

The API shall expose versioned, documented contracts for authentication, customers, vehicles, providers, bookings, dispatch, jobs, estimates, invoices, payments, notifications, support and administration. Write operations shall define validation, authorization, errors and idempotency where relevant.

Recommended conventions: JSON over HTTPS, consistent status codes, request IDs, pagination, filtering/sorting, ISO-8601 timestamps, structured errors, API versioning and OpenAPI documentation. Realtime connections shall be authenticated and server-authoritative.

11. Reporting & Analytics

- Operational metrics: active requests, matching rate, response time, ETA, completion and cancellation.
- Business metrics: request volume, repeat usage, ratings and provider performance.
- Financial metrics: customer charges, platform revenue, provider earnings, refunds and payouts.
- Analytics events shall use stable names and avoid unnecessary sensitive data.

12. Testing & Acceptance

- Unit tests for domain rules and critical utilities.
- Integration tests for database, authentication, dispatch, payments and notifications.
- End-to-end tests for the complete customer-provider journey.
- Authorization tests for every role and ownership boundary.
- Failure-path tests for retries, duplicates, payment callbacks, cancellations and network loss.
- Load/performance tests for request creation, matching, tracking and admin queries.
- Security testing including dependency scanning, secret detection and API abuse/rate-limit testing.
- Critical acceptance criteria must pass before production release.

13. Deployment & Environment Requirements

- Separate development, staging and production environments.
- Environment-driven configuration and secure secret management.
- Containerized services and automated CI/CD where appropriate.
- Controlled database migrations.
- Health/readiness checks and graceful shutdown.
- Rollback strategy and tested backup/recovery procedures.
- Centralized logs, metrics and alerts.

14. MVP Acceptance Criteria

Area	Minimum acceptance
Customer	Authenticate, manage vehicle, request assistance, receive provider, track, approve estimate, record
Provider	Register, submit verification, configure services/availability, receive/accept request, update job, est
Admin	Authenticate, verify providers, monitor jobs, manage users, inspect payments and handle support/c
Backend	Critical state transitions and permissions enforced server-side with persistent history/audit.
Realtime	Active status/location updates work with stale/offline handling.
Payments	Success, failure, duplicate and delayed-payment cases handled safely.
Security	No critical authorization bypass, secret exposure or unsafe upload path.
Operations	Deployment, backup, monitoring and recovery procedures documented and tested.

15. Requirements Traceability Baseline

Domain	Primary artifact
Authentication & RBAC	Backend schema + API specification + security architecture
Customer & vehicle	Database schema + API specification + UI/UX specification
Matching & dispatch	Backend architecture + realtime architecture + implementation plan
Booking lifecycle	Backend schema + state machine + API specification
Inspection & estimates	Job schema + UI/UX + API specification
Billing & payment	Payment specification + database schema + security
Tracking	Realtime architecture + maps integration
Admin & support	Admin UX + API + operations SOP
Quality	QA plan + observability + deployment documentation

16. Future Extensibility

- Support scheduled service alongside emergency assistance.
- Support individual mechanics and multi-user garages.
- Reuse vehicle/service history for future maintenance intelligence.
- Allow configurable matching so advanced scoring/AI can be introduced independently.
- Support additional payment methods, refunds and provider settlement.
- Extend toward fleet accounts, EV services, parts, insurance and partner APIs.

17. Definition of Done

- Requirement has a unique ID and acceptance criteria.
- UI, API and database behavior are aligned.
- Authorization and failure paths are implemented.
- Automated tests cover normal and critical edge cases.
- Operational logs/metrics exist for important behavior.
- Documentation is updated with implementation changes.
- Feature passes staging acceptance before production release.

18. Final SRS Summary

RoadResQ is specified as a production-oriented platform connecting vehicle owners with verified automotive service providers. The system is centered on a server-authoritative booking lifecycle, capability-aware matching, realtime operational visibility, controlled estimates and approvals, auditable billing/payment records and role-based administration. This SRS is the requirements baseline; architecture, API, database, UX, security, testing and deployment documents refine implementation details.